INTERIM FEDERAL SPECIFICATION

LIGHT, EXTENSION, ELECTRICAL, FLUORESCENT

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Washington 25, D.C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This Interim Federal Specification covers portable electric fluorescent-lamp extension lights, for rough service use, complete with lamp, starter, ballast, connector plug, cord, lampholder, reflector, shield, handle, and end cap-hanger assembly. These lights are not for use in explosion-hazard areas.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids form a part of this Interim Federal Specification to the extent specified herein.

Federal Specifications:

W-C-596 - Connector, Plug, Electrical; Connector, Receptacle, Electrical; Plate Wall, Electrical
W-L-116 - Lamps, Fluorescent
W-S-755 - Starter, Fluorescent Lamp
PPP-T-45 - Tape, Gummed, Paper, Reinforced
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Fiberboard

Federal Standards:

Fed.Std.No.102 - Preservation, Packaging, and Packing Levels.
Fed.Std.No.123 - Marking for Domestic Shipment(Civilian Agencies).

(FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE).

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Military Specifications:
MIL-L-10547 - Liners, Case, Waterproof.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Official Classification Committee Publication:

Uniform Freight Classification.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, 33rd Street, New York 16, N.Y.).

Underwriters' Laboratories, Inc., Standards:
UL No. 542 - Lampholders, Starters, and Starter Holders for Fluorescent Lamps.
UL No. 935 - Electric-Discharge Lamp Accessory Equipment.
UL No. 62 - Flexible Cord and Fixture Wire.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, New York; or 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Blvd., Santa Clara, California.)

3. REQUIREMENTS

3.1 Description. The extension light shall be a complete unit, including connector plug, cord, handle, starter, ballast, fluorescent lamp, shield and an end cap-hanger assembly.

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3.2 The extension light shall be designed and constructed to withstand the strains, jars, vibrations, adverse weather and other similar conditions incident to shipping, storage, and service intended. All parts subject to wear shall be readily accessible for adjustment or repair. The extension lights shall be designed for 115-volt operation, but shall be capable of functioning at a maximum of 250 volts.

3.3 Construction. Construction shall be free from any characteristics or defects that may render the extension light unsuitable or inefficient for the intended purpose.

3.3.1 Fastening devices. All fastening parts shall be installed with adequate means for preventing loss of tightness and adjustment. Parts subject to removal shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.2 Electrical connections. Electrical connections shall be either mechanically secured, soldered, or brazed. Use of soldering acids, acid fluxes, or soldering salts shall be avoided, whenever practicable. Where used, all joints or terminals shall be thoroughly cleaned and all traces of acid removed.

3.3.3 Interchangeability. All replaceable parts shall be constructed to definite standards, tolerances, and clearances in order that any such part of a particular model may be replaced without requiring modification. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number.

3.4 Fire and casualty hazards.

3.4.1 Each bidder shall submit to the contracting agency proof that the fluorescent lamp extension light he proposes to supply under this specification conforms to all the applicable requirements of Underwriters' Laboratories, Inc. Standards, including Standards UL 62, 542, and 935. The label or reexamination listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the fluorescent lamp extension light conforms to these requirements.

3.4.2 In lieu of the label or reexamination listing, the bidder may submit independent proof satisfactory to the contracting agency that the fluorescent lamp extension light conforms to the requirements, including methods of tests, of Underwriters' Laboratories Standards.

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3.4.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his fluorescent lamp extension light.

3.5 Connector plug. Unless otherwise specified, the plug shall conform to W-C-596, styles D23, D24, D25 (2 pole, 3 wire grounding type).

3.6 Cord. The cord shall conform to UL No. 62 Standard for Flexible Cord and Fixture Wire. The lengths of the exposed cord between the plug and the handle assembly shall be that specified, with the following tolerances:

25 feet - plus 3 inches, minus 0 inches
35 feet - plus 3 inches, minus 0 inches
50 feet - plus 6 inches, minus 0 inches
100 feet - plus 12 inches, minus 0 inches

The cord shall be type SO. No. 16 AWG, 3 conductor.

3.7 Handle assembly. The handle assembly shall house the starter and lampholder, and shall provide a suitable groove or other means of support for the shield assembly. Handles shall be so constructed as to permit, without the use of special tools, removal and replacement of fluorescent lamp, and all other parts without damage. The handle shall be made of a rubber or other material equivalent to the jacket of the cord. The handle shall be of a size and shape as to permit full one-hand grasping for ease of carrying and operating the extension light unit. Cord, fittings, and lampholder openings shall be weatherproof with cord and lamp in place.

3.7.1 End cap, hanger assembly. The extension light unit shall be provided with a sturdy hangup hook, mounted in the end cap. The end cap shall be fabricated of the same material as the handle. The hook shall be constructed of not less than No. 7 BWG and shall be covered with a permanent coating of an insulating material not less than 1/32 inch thick, free from blisters, cracks or checks. It may be of the fixed or rotating type. If the hook rotates, provision shall be made to prevent chafing of the hook coating in turning by use of a molded grommet or other suitable means. The free end of the hook shall extend and shall be so inclined that the light can be suspended safely from a round horizontal support of a diameter up to that of the hook opening.

3.8 Lampholder. The lampholders shall be of molded composition or of metal with molded or fabricated composition insulation. The material shall not distort or soften at 250 degrees F. Lampholders shall hold the lamps securely without the use of auxiliary devices.

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3.9 Starter socket. Starter sockets where required, shall be designed so that starters may be removed from the socket by the fingers without the use of tools.

3.10 Starter. Starters shall meet the applicable requirements of W-S-755.

3.11 Fluorescent lamp. The fluorescent lamp shall conform to the applicable requirements of W-L-116. The lamp shall be adequately shielded to prevent any possibility of injury to the user, and shall be provided with a suitable reflector, to permit easy focusing of the light on the work.

3.12 Identification data. The handle assembly shall show the manufacturer's name or trademark and model number permanently inscribed on or attached thereto.

3.13 Performance.

3.13.1 Resistance to shock. The extension light unit with the lamp lighted, shall not suffer any visible injury or damage to the electrical circuit when tested in accordance with 4.4.1.

3.13.2 Resistance to the entrance of moisture. The extension light unit shall meet the test of 4.4.2 without the entrance of any moisture.

3.13.3 Dielectric strength. Extension lights shall withstand a potential of 1250 volts, 60 cycles, between any two conductors or between any conductor and other exposed metal parts, without damage or breakdown when tested in accordance with 4.4.3.

3.13.4 Strain relief. Strain relief shall be provided to prevent mechanical strain of a direct pull on the cord from being transmitted to cord terminals, causing breaking or distortion of the terminal screws or the cord ending when tested in accordance with 4.4.4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

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4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All extension lights of the same type presented at one time shall be considered a lot for purposes of inspection.

4.3 Examination. The sample for visual and dimensional examination shall be in accordance with MIL-STD-105. Each sample shall be given a general examination for workmanship, material, general design, assembly design where applicable, insulation, operation of switching mechanism where applicable, dimensions, finish, and markings, with a lot acceptance or rejection based on an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.4 Tests. The sample for tests shall be in accordance with inspection level L5 of MIL-STD-105 with reduced inspection procedure R-1 and with a lot rejection number of 1. The sample shall be tested to determine conformance to UL No. 542 Standard for Lampholders, Starters, and Starter Holder for Fluorescent Lamps, UL No. 935 Standard for Electric-Discharge Lamp Accessory Equipment and UL No. 62, Standard for Flexible Cord and Fixture Wire. Where requirements specified herein differ from or duplicate those of the UL Standards, the applicable test of UL need not be performed or duplicated.

4.4.1 Shock test. A sample shall consist of normally assembled extension light with a lamp in the socket. The assembly shall undergo low and high-temperature shock tests by dropping it in such a manner that the handle and shield assembly hit on the side.

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4.4.1.1 Low-temperature shock test. The sample shall be placed in an ambient temperature of minus 25 degrees Fahrenheit (F) for a period of 4 hours. Immediately upon removal, the sample shall be dropped twice onto a hard wooden floor from a distance of 5 feet. During the drop portion of the test, the lamp shall be burning.

4.4.1.2 High-temperature shock test. The sample shall undergo the test specified in 4.4.1.1 except that the ambient temperature shall be 160 degrees F.

4.4.2 Moisture test. The sample extension light shall have all parts disassembled from the handle assembly including cord, strain relief, and shields and reassembled three times prior to performing this test. With the lamp burning, the sample extension light as normally assembled shall be suspended by the hook, and subjected for one hour to a water spray, adjusted to be approximately the equivalent of a beating rain. The spray shall be applied to the top and sides of the light. At the end of the spraying, the exterior of the light shall be wiped dry and immediately subjected to the applicable dielectric strength test (see 4.4.3). The interior of the handle assembly shall be examined for the presence of water.

4.4.3 Dielectric strength test. One test lead shall be attached to the guard hook. The insulation shall be removed from the end of the hook only sufficiently to attach one test lead. The plug contacts shall be connected to one another, and the second high-voltage lead shall be applied to the plug terminals. The test voltage shall also be applied to adjacent conductors. The test voltage specified in 3.13.3 shall be applied for one minute.

4.4.4 Strain relief test. The sample shall be an assembled extension light. The test load shall be 75 pounds. The test load shall be applied slowly and steadily for a period of 10 minutes on the cord with the handle held in place. At no time during the test shall the terminal screws or the terminal ends of the cord take any of the load. At the completion of the test the extension cord shall be examined for conformance to 3.13.4.

4.5 Inspection for preparation for delivery. The preservation, packaging, package, and marking of the lights shall be inspected to determine compliance with section 5.

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5. PREPARATION FOR DELIVERY

(For civil agency procurements, the definitions and applications of the levels for packaging and packing shall be in accordance with FedStd.No.102.

5.1 Packaging.

5.1.1 Levels A. and B. Extension lights shall be individually packaged in fiberboard containers conforming to PPP-B-636, type I or II, class 1, style RSC minimum bursting strength of 200 p.s.i. The cord shall be rolled or folded in a compact manner and sufficient cushioning shall be provided to prevent movement and damage during handling and shipment. Each container shall be closed in a manner to prevent accidental opening.

5.1.2 Level C. Extension lights shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 Level A. Extension lights packaged as specified in 5.1.1, shall be packed in any of the following containers at the option of the contractor:

Specification	Type and Class
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2

Containers shall be provided with caseliners conforming to MIL-L-10547, closed and sealed in accordance with the appendix thereto. Caseliners may be omitted for fiber boxes conforming to PPP-B-636 provided all corners, edge seams and manufacturer's joints are sealed with minimum 1-1/2 inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes may be banded with filament reinforced tape in accordance with the appendix to PPP-B-636. The gross weight of wood or wood cleated boxes shall not exceed 150 pounds, of fiberboard boxes 70 pounds.

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5.2.2 Level B. Twelve (12) Extension lights packaged as specified in 5.1.1, shall be packed in a fiberboard box conforming to PPP-B-636, type I or II, class 1, style RSC, bursting strength of not less than 275 p.s.i. Closure shall be adhesive applied over at least 75 percent of the flap area or by the two strip tape method as specified in Rule 41, section 7, of Uniform Freight Classification, using tape conforming to PPP-T-45.

5.2.3 Level C. Extension lights shall be packed in accordance with the supplier's standard commercial practice in containers conforming to carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military activities. Marking of interior packages and shipping containers shall be in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking of interior packages and shipping containers shall be in accordance with Fed.Std.No.123.

5.3.3 Special markings. Each shipping container shall be imprinted as follows: "Glass - DO NOT DROP OR THROW." Letters shall be conspicuously positioned and shall be minimum of 2 inches in height, space permitting.

6. NOTES

6.1 Intended use. The extension light is for the purpose of furnishing a safe, fluorescent light source at the end of a long cord where necessary, for work in and around shops, garages, normally unlighted, and inaccessible areas. These lights are for rough-service use where the cord is apt to be subjected to abuse and wear such as from trucks, packing cases, and oil and also where there may be rain or very moist conditions.

6.2 Ordering data. Purchasers should exercise any desired options contained herein and should specify the following in procurement documents:

(a) Title, number, and date of this specification.
(b) Length of cord required (see 3.6).
(c) Plug requirement, when not as specified (see 3.5).
(d) Length of time inspection records are to be kept (see 4.1).
(e) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

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6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, and lengths of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover the kinds which are suitable for Federal Government requirements.

6.5 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:
Electrical cord sets
Carload minimum weight 30,000 pounds

Motor:
Electrical cord sets, No. 1
Truckload minimum weight 30.6
pounds subject to Rule 34, National Motor Freight Classification

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